ECE 329 **Homework 12** **Due: Fri, Dec 19, 2025, 11:59PM**

5 problems totaling 30pts with 35pts extra Bonus Points. (Bonus problems are optional and the earned bonus credits will be used to "fill in" the "missing points" in the total homework score.)

1. (10pts) **Bounce diagram problem:** Consider a T.L. with **characteristic impedance** $Z_o = 50\,\Omega$, length $l = 300$ m, and **propagation velocity** $v = \frac{1}{\sqrt{\mathcal{LC}}} = c = 3 \times 10^8$ m/s. A **voltage source** $f(t)$ with an **internal resistance** $R_g = Z_o$ is connected to one end of the T.L. (at $z = 0$) and the other end ($z = l$) is terminated by a **load resistance** $R_L = Z_o/3$.

   a) (2.5pts) Construct a "bounce diagram" to determine the voltage, $V(z,t)$, and current, $I(z,t)$, variations on the line for $0 < z < l$ and $t > 0$ for $f(t) = \delta(t)$. **Hint:** first construct the voltage diagram and then utilize it to form the current bounce diagram by division with Z_o and negation of voltages of back propagating waves (from load to the generator).

   b) (2.5pts) Write your expressions for $V(\frac{l}{2}, t)$ and $I(\frac{l}{2}, t)$ as weighted sums of appropriately delayed impulses $\delta(t)$.

   c) (2.5pts) Plot $V(\frac{l}{2}, t)$ as a function of t for $0 < t < 6\,\mu$s if $f(t) = 10u(t)$ V. **Hint:** use the convolution of the result of part (b) with $10u(t)$.

   d) (2.5pts) Determine the steady state voltage and current on the line for the excitation from part (c). **Hint:** With $10u(t)$, the voltage and response on the line should approach a DC state as $t \to \infty$ at which stage the transmission line can be treated as wires in a lumped element circuit!

2. **Bonus problem:** (15pts) Consider a parallel plate and air-filled transmission line with a plate separation of $d = 1$ cm in x direction and width W=20 cm in y direction such that weak fringe fields outside the region bordered by the plates can be ignored. In between the plates it is observed that

$$\mathbf{E}(z,t) = 4\cos(\omega(t - \frac{z}{c}))\hat{x}\,\frac{\mathrm{V}}{\mathrm{m}}.$$

a) (5pts) Determine $V(0,t)$ by computing $\int_{x=0}^{d} \mathbf{E}(z,t)\cdot\hat{x}dx$.

b) (5pts) Is $V(0,t)$ also the same as $\int_{x=0}^{d} \mathbf{E}(z,t)\cdot d\mathbf{l}$ if along the integration path $d\mathbf{l} = (\hat{x}+\hat{y})dx$.

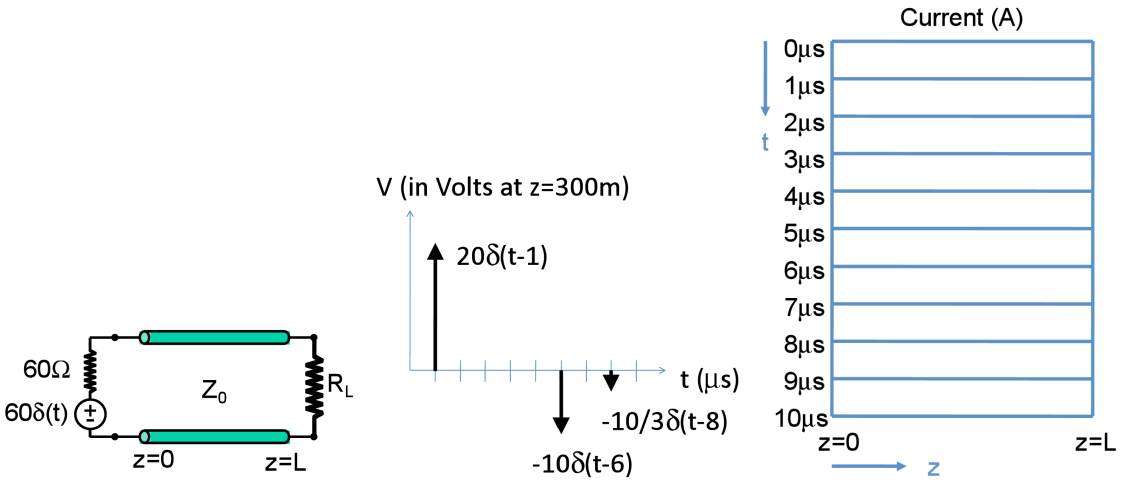
c) (5pts) If a voltmeter is used to measure the voltage drop V_{AB} from $A = (0,0,0)$ to $B = (d,0,\lambda/4)$, it will indicate $V_{AB}(t) = \int_A^B \mathbf{E}(z,t)\cdot d\mathbf{l}$ which is a line integral taken along the path formed by the wire leads of the voltmeter. Here λ is the wavelength of $\mathbf{E}(z,t)$ given by $\lambda = 2\pi/\beta$ where $\beta = \omega/c$. Evaluate $V_{AB}(t)$ for

i. (2.5pts) integration path going from A to $P = (d,0,0)$ and then from P to B along straight line segments;

ii. (2.5pts) integration path going from A to $Q = (0,0,\lambda/4)$ and then from Q to B along straight line segments;

and comment on why different functions $V_{AB}(t)$ are obtained in (i) and (ii).

3. (10pts) A voltage generator with internal resistance $R_g = 60\,\Omega$ and an open circuit output voltage $f(t) = 60\delta(t)$ feeds a T.L. (see the ckt diagram on the left below) that has an unknown characteristic impedance Z_o and an unknown resistive load termination R_L at an unknown distance, L, from the generator. At a distance $z = 300$ m from the generator, smaller than L, the voltage waveform as a function of time is as plotted below on the right.



a) (2pts) Determine the impedance of the transmission line, Z_o.

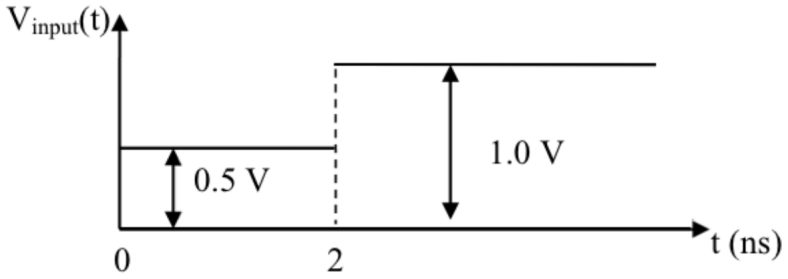
b) (2pts) Determine the load reflection coefficient Γ_L and the load resistance, R_L.

c) (2pts) Determine the length of the transmission line, L in meters.

d) (2pts) Sketch a bounce diagram for the *current* waveform $I(z,t)$ for $0 < t < 10\,\mu$s. Be sure to mark the numerical values for the amplitude of the current in the diagram.

e) (2pts) What is the algebraic expression for the current waveform as a function of (z,t) for the domain $0 < z < L$ and $0 < t < 10\,\mu$s.

4. (10pts) One of the wires of the transmission line is defective at some distance d away from its input point ($z = 0$). The defect manifests itself as an effective series resistance R_d at a distance d from the input. The transmission line is lossless, of length 2 m, and propagation speed on the transmission line is $v = 2 \times 10^8$ m/s. The transmission line is terminated at $z = 2$ m with a load resistance of $R_L = 100\,\Omega$. The voltage response observed at the input terminals of the defective and loaded 2 m line, when it is driven by a voltage generator with a unit step open circuit voltage $V_g(t) = u(t)$ V and an internal resistance of $R_g = 100\,\Omega$, is as depicted in the figure below:



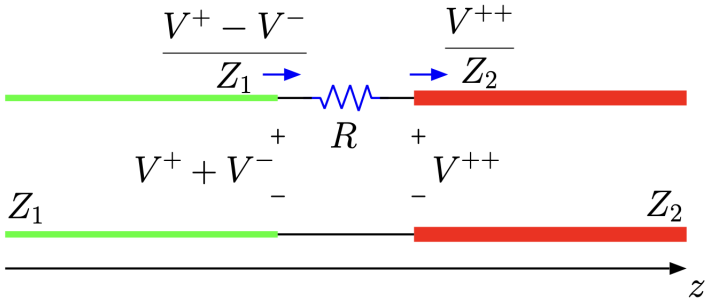
a) (2.5pts) Calculate the characteristic impedance Z_o of the transmission line. **Hint:** Use $V_{input}(0) = 0.5$ V.

b) (2.5pts) Calculate the distance d from generator where the defect is located. **Hint:** input voltage response to the unit step source is unchanging until $t = 2$ ns!

c) (2.5pts) Calculate the defect resistance R_d.

d) (2.5pts) Plot the voltage response at the load. **Hint:** almost no work here!

5. **Bonus Problem (20pts):** Two T.L.'s with characteristic impedances Z_1 and Z_2 are joined at a junction that also includes a series resistance R as shown in the diagram below.



a) (5pts) Write the pertinent KVL and KCL equations at the junction that relate $V^+(t - \frac{z}{v_{p1}})$, $V^-(t + \frac{z}{v_{p1}})$ and $V^{++}(t - \frac{z}{v_{p2}})$.

b) (5pts) Solve the KVL and KCL equations above to obtain the reflection and transmission coefficients $\Gamma_{12} \equiv \frac{V^-}{V^+}$ and $\tau_{12} = \frac{V^{++}}{V^+}$ for the junction.

c) (5pts) Calculate $\Gamma_{12} \equiv \frac{V^-}{V^+}$ and $\tau_{12} = \frac{V^{++}}{V^+}$ for $Z_1 = 100\,\Omega$, $Z_2 = 50\,\Omega$, and $R = 25\,\Omega$.

d) (5pts) Construct a voltage bounce diagram for this system up to $t = 8\,\mu s$ if both lines are 300 m long, both have free space propagation speed of 300 m/μs, Z_2-line is open circuit terminated as shown, and the input terminal of Z_1-line is an ideal voltage source $V_s(t) = \delta(t)$ with zero internal resistance. It is sufficient to indicate the impulse coefficients on the bounce diagram.